

Representation Convergence - Status at a Glance

All models pretrained and frozen; only closed-form linear maps fitted, except rows marked (trained). Thesis: independently trained models share representations up to a linear transformation - and that convergence is deployable.

Exp	Models	Input data (modality; train / eval split)	Target	Achievement
A	GPT-2 vs Pythia-160M	TEXT only - 10k WikiText passages, no labels or captions. Fit 7,500 (9.8 rows/dim) / eval 2,500 held out, 75:25 split	Do two independently trained LLMs converge?	Yes, linearly: stitching R2 0.74-0.80 vs 0.17 random control; CKA 5.3x control; correspondence real but hub-dominated; eff. dim 2.2-10.9 vs 9.3-92.7 of 768 explains the metric artifacts
B	MobileCLIP-S1 -> SigLIP 2	IMAGES only for the fit - 4k COCO val2017, each encoded twice; captions enter only at evaluation. Fit 3,000 (5.9 rows/dim, bottom of band) / eval 1,000	Are two vision encoders linearly interchangeable?	Ridge R2 0.592 / cosine 0.900; Procrustes fails (0.038) -> anisotropy (69x spread in W's singular values); MLP +0.003 -> the relationship is exactly linear
B3/D	same + Qdrant index	IMAGE + TEXT - no fitting. Eval 1,000 : the B held-out images as the gallery, their first captions as queries (gallery size 1,000 throughout)	Is the convergence deployable?	One 0.79 MB fp16 adapter, one shared index: R@5 = 95.1% of ceiling; raw = chance; free extra capability: cross-tier membership R@1 0.987
E1	DINOv2 -> bge-m3	IMAGE-CAPTION pairs - 8k COCO train2017. All 5 captions (what COCO provides) averaged per image: cancels annotator idiosyncrasy, R2 0.278 -> 0.511. Fit 7,000 (9.1 rows/dim) / eval 1,000 ; ceiling re-measured under the same averaging (0.648 -> 0.875)	Does convergence cross modalities without joint training? (PRH)	Measurable but weak: R2 0.511; R@1 0.358 = 40.9% of a matched SigLIP ceiling, 358x chance; verdict robust across caption protocols; cross-modal costs ~half of same-modality R@1
E1 sweep	DINOv2 small / base / large -> bge-m3	IMAGE-CAPTION pairs , same protocol and cls+patch pooling at every point. 9.5k / 18k / 23.8k pairs - N scaled with output width so rows/dim is 11.1 throughout; eval 1,000 each, ceiling re-measured per run	Does alignment grow with encoder capacity? (the PRH trend)	Yes - measured. R@1 = 40.9 / 53.5 / 58.3% of a matched ceiling across 13.8x params; monotone in R2 (0.516/0.580/0.609), cosine, all three recalls AND the shuffle gap (0.616/0.682/0.709). +15.3 points of ceiling per decade (fit R2 0.949); base and large exceed the same-modality 0.592. Returns are compressive - no extrapolation beyond 304M
E1.1	DINOv2 -> GPT-2 (vs bge-m3, matched pair)	IMAGE-CAPTION pairs , identical to E1 by construction. Fit 7,000 (9.1 rows/dim) / eval 1,000	Does the text side need similarity training?	Predicted split: R2 retains 74%, retrieval 27%; GPT-2 space is near-1D (eff. rank 6.5/768, pair-cos +0.999) so cosine is degenerate (reads 1.000 at 9.5% retrieval) - metric validity depends on the space's geometry
E1.2	DINOv2 -> GPT-2 + label-free corrections	no new data - reuses the E1.1 embeddings (base, cls+patch). Transforms fitted on the 17,066 train targets only; eval 1,000	Is the collapse missing correspondence, or missing isotropy?	Isotropy. Whitening - unsupervised, label-free - lifts R@1 0.130 -> 0.478 , reaching 102.6% of the contrastively trained bge-m3 arm. Effective rank vs R@1 tracks monotonically (rho +0.94). Centering alone fails (fixes pair-cosine, not spread). Whitened R2 is 0.014 with the best retrieval of any arm - the metrics fully decouple
E2/E3	the E1 stack + SigLIP text tower	E2 demo: 500 unseen val2017 images + captions as the gallery. E3: eval 1,000 (the B3 rows), plus a 200-3,000 row data-efficiency curve	Demo both search directions; can a consumer USE the map?	Working two-way demo; stitching penalty R@5 = 0.049 -> the unchanged SigLIP text tower consumes translated vectors near-transparently
F1-F2	SigLIP 2 teacher, MLP head (trained)	IMAGES only, NO captions - the teacher supplies the target. Fit 38,000 (74 rows/dim; MLP 0.029 examples/param) / eval 2,000 in-corpus, plus the 1,000 B3 rows for comparable retrieval	Was the +0.003 MLP control data-starved?	No: at 74 rows/input-dim the MLP converges to EXACTLY the linear solution (0.9052 = 0.9052) - bottleneck confirmed twice
C1.1	MobileCLIP <-> SigLIP 2, no pairing	IMAGES only , 1,500 vectors per side. The pairing is destroyed and the true permutation sealed until scoring; eval on the standard 1,000 rows	Can the correspondence be found with NO paired examples?	Partially, yes. Gromov-Wasserstein on shape alone recovers 39.7% exact matches (595x chance); the map fitted from it reaches R@1 0.252 = 40% of ceiling, verification AUC 0.906 - with zero pairs. Two measured negatives: ICP degrades it, and whitening (which rescues retrieval in E1.2) destroys it - isotropy helps you READ a correspondence, hurts you when you must FIND one
G1	5 encoders -> one 512-d hub	5 cached spaces , same 9,533 images: DINOv2 small/base/large, bge-m3, GPT-2. Balanced whitened-PCA hub; one linear map each; 8,533 train / eval 1,000 ; width x alpha swept, rows/dim >= 5 enforced	Is there ONE shared coordinate system, usable across encoders?	Yes - demonstrated by use. A caption head trained on ONE encoder's hub vectors reaches 95.9% / 92.9% of native on two encoders it NEVER saw (absolute R@1 rises 0.398 -> 0.468); random-map control at chance; verif AUC 0.98. Transfer cliff at 512 -> 768 dims (whitened-PCA noise amplification); portability and lossless reconstruction pull hub width in opposite directions - the C.11/C.12 tension in a third form
E1.3	encoder ladder: GPT-2 / BERT / SBERT / bge	same image cache (DINOv2-base cls+patch, 18k pairs, 11.1 rows/dim); only the TEXT objective varies. Predictions pre-registered before running	Does C.11's mechanism predict encoders it never saw?	P1 confirmed exactly: raw R@1 orders by eff.rank, rank-corr +1.00 (6.5/65/93 -> 0.160/0.362/0.510); BERT lands between GPT-2 and SBERT as registered. P2 partial - the sharper finding: whitening lifts every arm (0.478/0.521/0.588) but a 0.110 spread remains in raw order: isotropy is necessary, not sufficient - contrastive training also arranges WHICH directions carry meaning, and whitening cannot install that
D.2	adapter as a threshold decider	1,000 held-out images , the shipped adapter scored as three decision rules (verification / ranking / membership)	How well does one map decide, not just rank?	A hierarchy. Verification AUC 0.999 , per-query ranking R@1 0.987 , membership under one global cutoff AUC 0.889 (best F1 0.840). Harder as the decision gets more global (best-of-500 impostor). ROC fits thresholded tasks, not R@K; AUC measures separation not magnitude (caption-verif AUC 0.997 at cosine 0.136)
F5	refit linear W (38k rows)	Fit images only (38,000). Eval 5,000 image-caption pairs from train2017 beyond the corpus - 5 disjoint 1,000-galleries, unseen by both maps (5x the usual eval, to settle R@1)	Is the shipped adapter improvable by data alone?	Yes, significantly: R@1 +0.024, R@5 +0.012, R@10 +0.011 on 5,000 fresh paired queries (all CIs exclude 0); same artifact, same seconds-long refit; 90%-of-ceiling gate approached (89.0%), not met
F3-F4	MobileCLIP last stage (trainable)	IMAGES only , same 38k corpus restreamed (the trunk trains). Fit 38,000 / eval 2,000 , then the 1,000 B3 rows	Can the ceiling itself move?	Built, deliberately gated: justified by F2's verdict but trades the seconds-refit property for full retraining

No training in A/B/E: all models frozen (no_grad); the only fitted object is a closed-form ridge matrix W outside both models; weights are bit-identical after every experiment. Captions never enter DINOv2; pixels never enter bge-m3 - the pairing is row-index bookkeeping. F-series rows train new artifacts beside the frozen encoders (F2: a separate head); F3, which would touch encoder weights, is built but unrun - every pretrained model remains bit-identical to its download.

Bottom line: convergence confirmed on three model pairs (text-text, vision-vision, vision-text); exactly linear; deviation from rigidity is measured anisotropy; shipped as a 0.79 MB adapter at 95% of ceiling and improved further by refitting on more unlabeled data. Sharpest methodological finding: similarity metrics are only interpretable after measuring the space's geometry - cosine read 1.000 in a space where retrieval ran at 9.5%.